Inorganic Cumulative Examination

November 18, 1995

You have just completed the synthesis and purification of two metal complexes and wish to characterize them. The complexes were made from cobalt chloride and phenylalanine (amino acid with $-CH_2-C_6H_5$ substituient). Complex I was prepared under anaerobic conditions and Complex II was prepared in air saturated solutions. As a starting hypothesis we will assume that complex I contains a Co(II) metal center since it was prepared under reducing conditions and complex II contains Co(III). You have the following spectroscopic techniques available to you: 1) UV/Vis 2) IR, 3) ^{1}H NMR 4) EPR and 5) C, H, N elemental analysis (Co:ligand ratios ?). In characterizing these compounds you wish to confirm the oxidation state of the cobalt center (2+ or 3+), the nature of the ligands around the metal and the atoms of the ligands which are bonded to the metal. Describe how you would address these issues with each of the analytical options available to you..